



# Sqrt Tree

Given an array  $a$  that contains  $n$  elements and the operation  $\circ$  that satisfies associative property:  $(x \circ y) \circ z = x \circ (y \circ z)$  is true for any  $x, y, z$ .

So, such operations as  $\gcd$ ,  $\min$ ,  $\max$ ,  $+$ ,  $\text{and}$ ,  $\text{or}$ ,  $\text{xor}$ , etc. satisfy these conditions.

Also we have some queries  $q(l, r)$ . For each query, we need to compute  $a_l \circ a_{l+1} \circ \dots \circ a_r$ .

Sqrt Tree can process such queries in  $O(1)$  time with  $O(n \cdot \log \log n)$  preprocessing time and  $O(n \cdot \log \log n)$  memory.

## Description

### Building sqrt decomposition

Let's make a [sqrt decomposition](#). We divide our array in  $\sqrt{n}$  blocks, each block has size  $\sqrt{n}$ . For each block, we compute:

1. Answers to the queries that lie in the block and begin at the beginning of the block ( $\text{prefixOp}$ )
2. Answers to the queries that lie in the block and end at the end of the block ( $\text{suffixOp}$ )

And we'll compute an additional array:

1.  $\text{between}_{i,j}$  (for  $i \leq j$ ) - answer to the query that begins at the start of block  $i$  and ends at the end of block  $j$ . Note that we have  $\sqrt{n}$  blocks, so the size of this array will be  $O(\sqrt{n}^2) = O(n)$ .

Let's see the example.

Let  $\circ$  be  $+$  (we calculate sum on a segment) and we have the following array  $a$ :

```
{1, 2, 3, 4, 5, 6, 7, 8, 9}
```

It will be divided onto three blocks: `{1, 2, 3}`, `{4, 5, 6}` and `{7, 8, 9}`.